

Radio spectrum for mobile networks

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The foundations of all mobile telecommunications networks are embedded in the radio spectrum to which the network operators have licensed access — without such spectrum there is no mobile business. In the changing world environment for telecommunications, and particularly within the increasingly liberalised markets of Europe, spectrum availability has become one of the key factors influencing progress. The availability of mobile spectrum is therefore, quite understandably, the subject of very dynamic political, regulatory, technical and market forces.

The BT Group, as a major UK user of spectrum for mobile and many other radio applications, remains proactive in securing and developing its spectrum interests. Furthermore, as BT increasingly seeks to explore overseas mobile (and indeed fixed) telecommunications markets, its understanding of the radio regulatory environment is being extended progressively to cover many other countries.

This paper reviews recent and current developments in those areas of the spectrum of greatest interest to the mobile operator, and looks at the main issues that have an impact on the growth of mobile service frequency allocations. Background information on the way in which the spectrum is managed at the national, European and global levels is also provided.

1. Introduction

Society now places a high priority on the economic, social and commercial benefits of public-access cellular radio, and there are constant pressures to extend the amount of spectrum available to such networks. The phenomenal growth of the cellular mobile market can only be sustained if more spectrum can be transferred to this role from other established applications. However, with the overall demand for VHF/UHF frequencies from all classes of spectrum users now far exceeding that which is available, competition for access to the relevant frequency bands is intense. New spectrum for public mobile telecommunications systems is indeed hard to find.

The most realistic possibilities lie in ‘re-farming’ the existing telecommunications frequency bands. The historical usage of some of the telecommunications frequencies therefore needs to be reviewed, with long-established ‘fixed’ radio applications being encouraged to give way to the new mobile networks. Such changes are indeed already underway, but the time-scales involved in revising frequency allocations and, more importantly, in actually removing or re-locating the established users, are protracted.

The delays in reorganising the spectrum are a source of obvious concern among those promoting liberalisation and those wishing to develop the mobile networks. Changes are therefore necessary to the way the spectrum is managed in order to get the telecommunications frequency bands in

appropriate shape for the future. It is possible, for instance, that some forms of (significant) spectrum pricing and/or disturbance compensation will have to be applied to telecommunications spectrum as devices for expediting change.

While in the past the organisation of the spectrum was largely a technical and radio regulatory matter, the drive towards liberalisation and competition, especially in mobile telecommunications, has added a significant political dimension to the management of those parts of the spectrum intended for telecommunications use.

With so much change underway, and particularly with 3rd generation¹ mobile systems (TGMS) concepts now beginning to stabilise (which will require significant spectrum ‘re-farming’ in the 2 GHz band), the coming few years will provide many important challenges for spectrum managers. It is therefore timely to reflect upon the spectrum allocations from an operator’s perspective, and to look at some of the main issues that will impact on the development of mobile telecommunications interests over the coming few years.

¹ The original analogue cellular networks (NMT, TACS, etc.) are considered to be 1st Generation systems. The current digital systems (GSM, etc.) with inter-network roaming and advanced network services are considered as the 2nd generation. Third generation systems will support a range of advanced (and wider bandwidth) services and true global roaming via both terrestrial and satellite elements (see also Groves and Clapton [1]).

2. The complex world of spectrum management

Spectrum management embraces a continuous and turbulent interaction between the spectrum requirements of fixed and mobile terrestrial services, fixed and mobile satellite networks, broadcasters, space scientists and radio astronomers, governmental and military networks and the numerous other smaller users. It is therefore necessary to maintain a broad understanding of what is happening in the spectrum context in order properly to identify opportunities for, and threats to, the telecommunications operators' radio-based commercial interests.

As noted above, changes in spectrum allocations are needed to allow ongoing growth of the mobile networks. Technical studies and formal management decisions leading to changes in spectrum use take place in many different fora. To become a credible player in the spectrum management community it is essential to establish expertise and experience in working with the ITU (International Telecommunications Union) Management Committees, World Radio Conferences and Study Groups, the European Commission (EC), the UK Department of Trade and Industry (DTI), the UK Radiocommunications Agency (RA), CEPT (European Conference of Posts and Telecommunications Administrations)/ERC (The European Radio Committee of the CEPT), ERO (The European Radio Office, a permanent organ of the ERC), ETSI (European Technical Standards Institute), the manufacturers (via liaison with FEI (UK Federation of Engineering Industries) and ECTEL (European Committee of Telecommunications Manufacturers)), and ETNO (European Public Telecommunications Network Operators Association). Each of these organisations has one or more (sometimes many) project teams, working groups and sub-groups involved in spectrum-related activities.

For any spectrum user to be effective in the pursuit of its interests, it will need to have constant awareness of the activity on the important issues within these many organisations, to identify when and where specific action is needed and to deliver carefully targeted and timely interventions in the appropriate fora. The task of co-ordinating mobile and other telecommunications spectrum requirements against the varied activities and initiatives in these many organisations requires effective information collation, good team-working and a clear strategy on spectrum requirements.

3. ITU world radio conferences

3.1 Role of the WRCs

The ultimate responsibility for defining the usage of all the radio spectrum resides with the biennial World Radio Conferences (WRCs) of the International Telecommunications Union. The WRC is the competent body for

affecting changes to frequency allocations and the various Radio Regulations at the global and regional levels. In the ITU context, a frequency allocation consists of the identification of a frequency band within which the operation of a particular ITU radio service is allowed.

The work of the WRCs is influenced by technical, regulatory and political considerations, with mobile radio issues always being high on the political agenda.

The essential frequency allocations, regulations procedures and data necessary for the effective management of the radio spectrum from 9 kHz to 275 GHz are set out in the ITU Radio Regulations [2], a 3-volume compendium having the status of a global international treaty among the 184 national administrations who comprise the formal Membership¹ of the ITU.

The most recent World Radio Conference, WRC-95, was held in Geneva during the Autumn of 1995, and comprised over 1500 delegates from ~140 countries. BT had two representatives at the conference, attending as part of the UK national delegation. Resulting from these WRC-95 activities, the ITU has recently published the 2-volume 'Final Acts' [3] of the conference, which represent a complete revision of the radio regulations to be considered for ratification by the ITU Member administrations. It should not, however, be assumed that all ITU Member nations have ratified the most recent revision of the Radio Regulations. As with many treaty texts, formal ratification can take some time within particular administrations. Where radio regulatory discussions take place between Members, it is generally on the basis of the most recent 'mutually ratified' versions of the relevant Radio Regulations, unless other bilateral arrangements are agreed for a specific discussion.

3.2 Influencing WRC decisions

Any proposals made at national or European level to extend the spectrum available to public mobile networks must fall within the framework of the frequency bands allocated to the mobile service from time to time by the WRCs. Getting the framework right at the ITU global/regional level is therefore essential.

WRC decisions are made by national administrations on a majority basis during the conferences, and where a vote becomes necessary it is on a one Member, one vote basis. Thus the USA and Japan have one vote each and the CEPT countries can (currently) bring together some 43 votes on issues of common interest.

¹ BT and numerous other companies and organisations are non-voting 'small m' members, but as such have no voice at the WRCs, etc.

Indeed, since the 1992 Conference, the CEPT countries have been co-ordinating their WRC interests via the CEPT Conference Preparatory Group (CPG), which endeavours to pull together European Common Proposals (ECPs) on the key issues (including issues relating to the mobile spectrum). CEPT does not itself have a formal voice in the ITU, and each ECP is therefore submitted to the ITU on a co-signature basis by the particular CEPT administrations that have endorsed it (not necessarily all 43 CEPT Members). Despite the co-ordinated European voice and the strong political influence of the US, the votes of the remaining majority, particularly the developing countries, can (and do) have a significant impact on the outcome of WRC decisions.

The telecommunications operators do not have a direct voice in WRC decisions, but by working co-operatively within the national, European and global spectrum management communities, and by helping to build constructive ECPs, they can assist in steering the development of mobile spectrum at the WRC level.

4. European spectrum management

4.1 *Telecommunications liberalisation and the drive towards spectrum harmonisation*

The European Union leads the world in striving for a fully liberalised telecommunications environment. Recent directives issued by the European Commission mean that, as from 1 January 1998, voice telephony, including mobile telephony, will be liberalised in all but a few ¹ of the EU member countries. Full liberalisation of non-voice services (i.e. data, multimedia, etc) will undoubtedly follow soon after.

Radio is seen as a major enabler of liberalised telecommunications markets. However, while mobile radio is at the focus of this activity, the EC vision also sees radio being used in many other areas of telecommunications networks to provide core infrastructure, mobile base station interconnections and to provide an alternative to copper and fibre for the access network. The speed at which a competitive network infrastructure can be built using radio is the key attraction. The European Commission is therefore keenly aware of the value of the spectrum in promoting competition and this has brought spectrum issues, especially those related to mobile networks, to the political and regulatory foreground. In the radio context, the EC telecommunications directorate (Directorate General XIII) must ensure that frequency allocations, equipment standards, licensing and type approval are all acceptable and applicable throughout the European Union, and preferably beyond its present boundaries.

A recent important event has been the amendment of the EU Mobile Directive so that operators can now have relatively unhindered access to the digital European cordless telephony (DECT) and DCS1800 frequency bands, etc, across Europe, subject only to the pre-requisites of the competition directives and to spectrum availability. This opening-up of the spectrum significantly eases the previously restrictive regulatory situation, and should allow many more mobile players to enter the European markets.

A cornerstone of the drive for European telecommunications liberalisation and the growth of competition is the policy of the EC to promote a consistent, i.e. harmonised, frequency table for the whole of Europe ². Equitable access to the frequency allocations within such a table, particularly for the established operators, is currently a key issue in relation to the furtherance of fair competition.

This goal of a European table of frequency allocations and utilisations (i.e. applications) is a major change of philosophy, as it makes the detailed management of frequency allocations (i.e. at the level below the global/regional allocations of the ITU) a primarily European rather than national responsibility (although national sovereignty is not compromised). However, this important change is supported by the EU countries because of the major economic, social, manufacturing and consumer benefits of a harmonised frequency table.

Nowhere are these benefits more openly visible than in the mobile radio context, where access to consistent advanced services (with GSM (global system for mobile) at least), as customers roam across Europe (and, indeed, many other countries of the world) with a single handset, is already whetting consumer appetites for 'global'-scale 3rd generation systems with even greater feature and service sets.

4.2 *The DSI process*

The decision was made by the ERC to consider the spectrum from 29.7 MHz to 105 GHz for European harmonisation by the year 2008. Once the harmonised European table was conceived, it was necessary for the ERC to gain an in-depth understanding of how the spectrum was currently being used within the CEPT community and, more importantly, to form a vision of how the spectrum should be re-farmed for the future. These tasks were given to the newly-formed European Radio Office, who initiated the CEPT-wide spectrum auditing and review process that became known as the Detailed Spectrum Investigation (DSI).

¹ Spain, Portugal, Greece and Eire will implement the directive somewhat later.

² It is not always clear whether, in the spectrum context, 'Europe' should be interpreted as the 15 EU Members countries, the 43 CEPT Member administrations or the defined geographical area of the continent. However, in this context, it is the CEPT membership.

There are six main stages of a DSI review.

- The ERC and ERO sets up a DSI management team comprising ERO and nominees from the broader European radio community (operators, manufacturers, users, etc). Terms of reference and objectives are determined by the ERC.
- The management team issues a call for submissions on the current and future use of the relevant frequency range from a wide variety of organisations and companies throughout Europe. The information requested is quantitative in terms of the numbers of systems of the various services currently deployed, and projected future growth, etc. The initial submissions are treated as commercially confidential and are not openly available.
- The DSI management team considers the responses within the guidelines of its terms of reference, and produces a detailed report and recommendations document for the ERC.
- The ERC considers the DSI findings and authorises publication of the report after requesting any necessary changes. The published report also includes an open invitation to comment to all interested parties.
- The DSI report, together with all comments received by the ERO, is passed to the CEPT-FM¹ and CEPT-SE² working groups with instructions to prepare detailed proposals for the harmonised European frequency table. The working groups are not bound by the DSI Recommendations, and experience has shown that considerable change can occur along the way.
- Once agreed proposals have been prepared by the working groups and approved by the ERC, they become the subject of an ERC Report or ERC Recommendation. (After issuing three Directives on GSM, DECT and DSRR (digital short-range radio) Spectrum, the EC delegated the responsibility for formal spectrum regulation to the ERC. This action was primarily to avoid conflicts in radio regulation between the EU members and the larger CEPT community. The ERC Decision therefore has virtually equivalent status to an EC Directive. The three earlier EC Directives were subsequently converted to ERC Decisions.)

¹ CEPT-FM — The ERC Working Group 'Frequency Management' which manages the European frequency table.

² CEPT-SE — The ERC Working Group 'Spectrum Engineering' which supports the work of CEPT/FM by examining the technical issues relating to spectrum management, i.e. compatibility, channel plans, spurious emissions, etc.

In 1992, when the DSI concept was formed, the CEPT-FM working group was nearing completion of a review of the 960 MHz to 3.4 GHz portion of the spectrum, initiated following the work of WRC-92, in preparation for the European implementation of 3rd generation mobile and satellite personal communications. The DSI study therefore avoided this frequency range, and split its review of the remainder into two phases.

The first phase DSI dealt with 3.4 GHz—105 GHz. This commenced in 1992, the DSI Phase 1 report was published in March 1993 [4] and the work was finally completed in March 1995 with the first formal publication of the European Frequency Table [5] by the ERC.

DSI phase II covered 29.7 MHz to 960 MHz and commenced in 1994. The deadline for the first round of submissions was November 1994. The management team completed its work in early 1995 with the 2nd phase DSI report [6] being approved and released by the ERC in March of that year. Comments on the Report and Recommendations were invited by September 1995.

This DSI covered an extremely important frequency range with major vested interests. Within this range are the professional mobile radio (PMR) frequencies, the UK analogue (TACS — Total Access Communications System) and digital (GSM) cellular frequencies, the (CT2 technology) cordless telephony frequencies and the proposed GSM extension bands. Also in this band are the UHF TV frequencies and Band II FM radio channels, and some important military communications bands.

Obviously there were proposals for additional PMR and cellular mobile spectrum, but these had to be evaluated against the broadcasters' own requirements for additional spectrum during the introduction of both Terrestrial Digital Audio Broadcasting (TDAB) and digital TV.

The DSI phase II committee finally produced 23 Recommendations, including a draft revision of the relevant section of the European Frequency Table, a sample of which (for 862—960 MHz) is shown in Table 1. It is worth noting that there are a number of features of this frequency table that contrast with the more familiar ITU table format (the ITU table is explained in the Appendix). These changes are quite significant.

Firstly, the frequency bands (column 1) are managed to a finer resolution when compared with the radio regulations (RR) region 1 allocations shown in column 2 of Table 1.

Next, the harmonised European Common Allocations (ECAs) in column 3 are more focused, i.e. the radio service

allocation intended to apply across Europe is defined as having Primary (protected) status (see the Appendix), and there are far fewer sharing options than is the case in the ITU table. However, the ECAs remain compatible with allocations in the ITU table.

An all-important additional column in the European table which differentiates its intentions from the ITU table, is the 'major utilisation'. This column defines the intended European harmonised applications, reinforced where appropriate by reference to CEPT/ERC decisions, which repre -

Table 1 Extracts from the DSI phase II Frequency Table for 862 MHz — 960 MHz.

Frequency band	RR allocation and relevant footnotes	European common allocation	Major utilisation	Notes
862.0000—875.0000 MHz	FIXED MOBILE except aeronautical mobile	MOBILE FIXED	SAB (services ancillary to broadcasting)	
875.0000—876.0000 MHz	BROADCASTING RR703	MOBILE	PMR	
876.0000—880.0000 MHz		MOBILE	PMR UIC (Union International de Chemin du Fer) Railway systems	If 4 MHz is not required for UIC then band available for other PMR applications
880.0000—890.0000 MHz	RR700B RR704	MOBILE Fixed	GSM Defence systems	
890.0000—915.0000 MHz	FIXED MOBILE except aeronautical mobile	MOBILE	GSM CEPT/ERC/DEC(94)01	Decision adopted by ERC in 1994
915.0000—920.0000 MHz	BROADCASTING RR703 Radiolocation	MOBILE METEOROLOGICAL AIDS Amateur Fixed	ISM (Industrial, Scientific and Medical, for RF-based, often high-powered, technical equipment) LPD (Low Powered Devices, deregulated devices such as burglar alarms, traffic light sensors, garage door openers, etc) 915—920 MHz Amateur 919.5—920 MHz New technologies 915—919.5 MHz	EU100 Narrowband windprofilers
920.0000—921.0000 MHz		MOBILE	PMR	
921.0000—925.0000 MHz		MOBILE	PMR UIC Railway systems	If 4 MHz is not required for UIC then band available for other PMR applications
925.0000—935.0000 MHz		MOBILE Fixed	GSM Defence systems	
935.0000—942.0000 MHz	RR704	MOBILE	GSM CEPT/ERC/DEC(94)01	Decision adopted by ERC in 1994
942.0000—960.0000 MHz	FIXED MOBILE except aeronautical mobile BROADCASTING RR703 RR704	MOBILE	GSM CEPT/ERC/DEC(94)01	Decision adopted by ERC in 1994

sent the principal use of the stated bands. In some cases, e.g. UIC (Union International de Chemin du Fer) Railway systems, the particular class of users is defined, in others (i.e. GSM), a particular European standard technology. It is likely that as the table develops, both the users and the standardised technologies will be defined for important bands. Overall, the European table sets down a far more prescriptive framework for the use of the spectrum to ensure that the European harmonisation goals are achieved.

The DSI recommendations that were of most importance to the public mobile telecommunications community are summarised as follows:

- abrogation of the ERC Decision TR20/10 which designated 2×2 MHz (888—890 MHz paired with 933—935 MHz) for DSRR systems — these unlikely-to-be-used allocations were preventing contiguous GSM expansion,
- deletion of the CTI cordless telephone allocations of 914—915 MHz and 959—960 MHz — these are not used in the UK, but overlap the GSM bands in some other countries,
- deletion of the specific reference to the earlier CT2 spectrum allocation from 864—868 MHz,
- closure of analogue cellular systems (at 450 MHz and 900 MHz) by the year 2008 at the latest,
- priority on the pan-European availability of the GSM extension bands of 880—890 MHz paired with 925—935 MHz,
- formalising of the ‘UIC’ bands as 876—880 MHz paired with 921—925 MHz, in anticipation that UIC will adopt a GSM-related technology,
- additional spectrum for PMR, principally for emergency service and civil trans-European trunked radio (TETRA) systems,
- possible additional UHF mobile allocations once the transition to digital TV broadcasting is completed.

4.3 CEPT-FM project team 25

Once the comments on the DSI Recommendations and frequency table had been collated (November 1995), the ERC set up a project team within the frequency management working groups (PT FM-25) to consider the DSI Recommendations and comments, and to finalise revisions of the European harmonised frequency table for the DSI/2 frequency range. The FM-25 project team is currently considering several possibilities for the 862—960 MHz frequency range to the ERC. One of the possible future scenarios is shown in Fig 1.

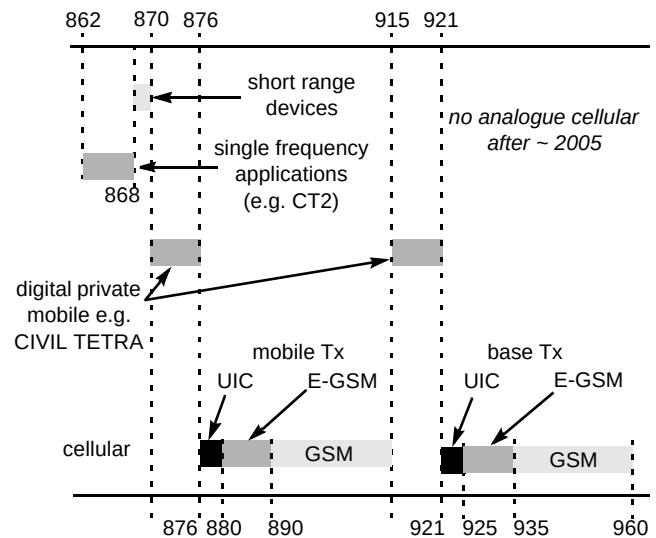


Fig 1 A possible future scenario for the 900 MHz band.

It can be seen from this example that, while these ideas are generally in line with the DSI Recommendations shown in Table 1, the CEPT project team has the option to go some way further in simplifying the usage of this important part of the spectrum. There will be further debate about these ideas within the CEPT groups, and Fig 1 does not necessarily represent the final outcome for this particular part of the band. Nevertheless, if this scenario were to be adopted, the mobile operators would benefit from a more certain spectrum environment at 900 MHz for the foreseeable future.

5. Mobile allocations at around 2 GHz

5.1 The general scenario at 2 GHz

The other key area of the spectrum of interest for mobile telecommunications development exists at around 2 GHz, and this has received considerable attention at recent WRCs.

It is in the bands around 2 GHz that the frequencies for personal communications services networks (PCN) (Mercury One-2-One and Orange in the UK), DECT and the future 3rd generation systems (including their satellite elements) are to be found, making this an important region of the spectrum which attracts significant political attention. Table 2 provides the relevant abstract from the ITU frequency table (as modified by WRC-95), and reveals the potentially complex sharing arrangements arising out of the mix of radio services allocated to these particular bands.

While primary mobile service allocations appear in all the frequency bands shown in Table 2, there are in fact several examples here of where potentially valuable mobile allocations are not realisable in practice because of sharing

difficulties. The following cases typify sharing problems that have to be accommodated.

- **Sharing between mobile and space science services**

Table 2 shows that the two paired bands 2025 — 2110 MHz and 2200 — 2290 MHz have allocations not only for the conventional fixed and mobile services, but also for some of the so-called ‘space science’ services, in this case the space operations, space research and earth exploration services. Space science systems have existed in these bands for some time. However, until WRC’92 their allocations had secondary (essentially unprotected) status. With the change to Primary status they are now entitled to full protection from future frequency assignments, including those for the terrestrial mobile service. It is generally accepted [7] that mobile cannot share with the space science services in these bands, and thus these potentially valuable ITU mobile allocations are effectively lost for the foreseeable future.

In the longer term it may be possible for the space science services to be moved to above 20 GHz [8], and it is likely that the initial stages of this process will be under discussion at WRC’97.

- **Sharing between fixed and mobile services**

In the telecommunications context, the fixed service generally supports telecommunications services with demanding performance and availability objectives, and thus the systems need to exist within a quantified and protected interference environment. Mobile systems have traditionally been far less demanding in this respect.

The protection normally required by stations of the fixed service make it very difficult for terrestrial mobile and terrestrial fixed telecommunications systems to share frequencies, at least within the same geographical area. Given that there is still a need for fixed service allocations at around 2 GHz (although this need is strongly contested by the mobile community), the bands have to be managed so as to avoid direct sharing between fixed and mobile in the same band. However, in this case, the social and political priority of mobile telecommunications seems to be prevailing, and it is the fixed service that is being encouraged to give way as far as is practicable..

Table 2 The ‘WRC-95’ Frequency Table at around 2 GHz.

1710—1930			FIXED S5.381 MOBILE S5.149 S5.341 S5.385 S5.386 S5.387 S5.388
1930 — 1970 FIXED MOBILE	1930 — 1970 FIXED MOBILE MOBILE-SATELLITE (Earth-to-space) S5.388	1930 — 1970 FIXED MOBILE	S5.388
1970 — 1980 FIXED MOBILE	1970 — 1980 FIXED MOBILE MOBILE-SATELLITE (Earth-to-space) S5.388	1970 — 1980 FIXED MOBILE	S5.388
1980 — 2010			FIXED MOBILE MOBILE-SATELLITE (Earth-to space) S5.388 S5.389A S5.389B S5.389F
2010 — 2025 FIXED MOBILE	2010 — 2025 FIXED MOBILE MOBILE-SATELLITE (Earth-to-space) S5.388 S5.389C S5.389D S5.389E	2010 — 2025 FIXED MOBILE	S5.388
2025 — 2110			SPACE OPERATION (Earth-to-space) (space-to-space) FIXED MOBILE S5.391 SPACE RESEARCH (Earth-to-space) (space-to-space) S5.392
2110— 2120			FIXED MOBILE SPACE RESEARCH (deep space) (Earth-to-space) S5.388
2120 — 2160 FIXED MOBILE	2120 — 2160 FIXED MOBILE Mobile-satellite (space-to-space) S5.388 S5.388	2120 — 2160 FIXED MOBILE	S5.388
2120 — 2170 FIXED MOBILE	2160 — 2170 FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) S5.388 S5.389C S5.389D S5.389E	2160 — 2170 FIXED MOBILE	S5.388 S5.388 S5.392A
2170 — 2200			FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) (space-to-space) S5.388 S5.389A S5.392A S5.389F
2200 — 2290			SPACE OPERATION (space-to-Earth) (space-to-space) EARTH EXPLORATION-SATELLITE (space-to-Earth) FIXED MOBILE S5.391 SPACE RESEARCH (space-to-Earth) (space-to-space) S5.392

There is a clear endorsement of the priority on mobile within Table 2, with the many references of footnote S5.388 in the various frequency bands; this footnote states: ‘The bands 1885—2025 MHz and 2110—2200 MHz are intended for use, on a worldwide basis, by administrations wishing to implement the future public land mobile telecommunication systems (FPLMTS). Such use does not preclude the use of these bands by other services to which these bands are allocated. The bands should be made available to FPLMTS in accordance with resolution 219 (Rev WRC-95)’.

Footnote S5.388 was originally introduced during the 1992 conference, and was perhaps the first instance of a WRC trying to venture beyond its traditional role of the allocation of radio services to frequency bands to suggest one particular (albeit generic) application for a band. This will undoubtedly become a growing trend for the future as the applications become global in nature.

The assumption must be, therefore, that the existing fixed service systems cannot continue to be used in this band in those countries implementing FPLMTS. Withdrawal of the existing fixed service systems (generally owned by PTOs or the military) will undoubtedly prove to be difficult in some countries. This is evident from similar problems in the smaller DECT and DCS1800 bands which, despite EC directives and ERC Decisions, are still not yet available on a pan-European basis; however, the complete vacating of these latter two bands has been successfully achieved in the UK.

- **Sharing between mobile and mobile-satellite services**

The problem that arises between the mobile service and the space science services also comes into play to a certain extent in connection with the mobile-satellite (MobSat) service. For 3rd generation mobile networks it is anticipated that there will be a mobile satellite component to ensure end-user access to true global roaming, the ideal being the provision of mobile telecommunications services to each and every point on the earth's surface, even in the most remote areas.

Power budget constraints for MobSat systems are such that the links to and from the satellites must operate with relatively low background interference levels. For the uplink (Earth-to-space) direction especially, it would not be possible for the MobSat channels to share frequencies with the vast numbers of terrestrial mobile connections. Whether or not the MobSat downlinks can co-exist (perhaps with indoor terrestrial mobile) remains to be determined. The development of spectrum for 3rd generation mobile systems will need to address the issues of compatibility between the terrestrial and satellite elements, with special provisions being made for the latter.

Figure 2 demonstrates the complexity of the current frequency usage in the 2 GHz band. This arrangement of

frequencies has come about in order to accommodate the sharing incompatibilities identified above.

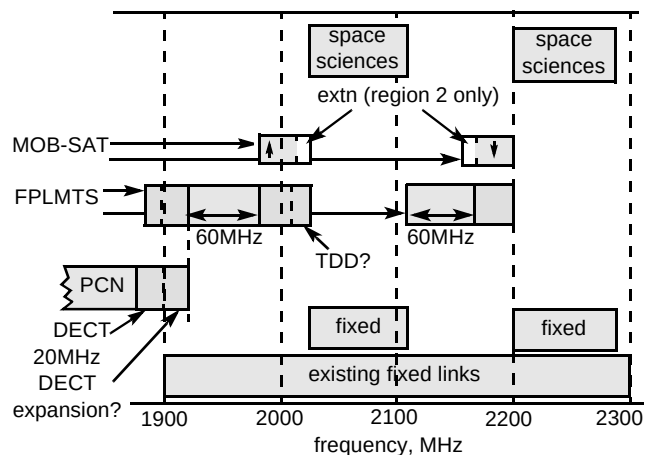


Fig 2 The emerging spectrum scenario for the 2 GHz band.

5.2 Third generation mobile spectrum

Despite the generally positive progress on spectrum development at around 2 GHz, concerns have arisen over the amount of spectrum available for terrestrial 3rd generation mobile systems.

The 'FPLMTS' bands of 1885—2025 MHz and 2110—2200 MHz total some 230 MHz, a figure originally conceived as being fully adequate to provide terrestrial 3rd generation mobile facilities, including 'wideband' multimedia services, to a truly mass market. However, a number of spectrum management decisions have eroded the amount of spectrum actually available for full duplex TGMS to the point where more spectrum is now seen as an essential requirement. These factors include the following.

- The mobile satellite bands of 1980—2110 MHz paired with 2170—2200 MHz. These consume 2×30 MHz, but the situation was compounded at WRC'95 by an extension of this spectrum (in the year 2005) in region 2 (the Americas) to 1980—2025 MHz paired with 2160—2200 MHz, as also shown in Fig 2. This further constrains the possibilities for truly global full duplex bands. The region 2 changes were needed to allow compatibility between the satellite bands and the now established US allocations for personal communications systems (PCS) which already occupy much of the FPLMTS bands.
- The 15 MHz between 1885—1900 MHz which already overlaps the European DECT frequency band of 1880—1900 MHz. DECT is likely to be successful and will undoubtedly dominate this band. The situation is compounded by the possibility of the extension of the DECT band upwards to 1920 MHz to provide

sufficient spectrum for all DECT interests, e.g. domestic cordless phones, cordless PABX systems, office LAN systems and wireless local loop.

- The general asymmetry of the FPLMTS bands, which leaves 15 MHz just above the lower satellite allocation unpaired in the higher sub-band. At the present time this oddment is being considered for (regulated or de-regulated) time division duplex (TDD) technologies because it is unsuited to the mainstream frequency division duplex application. The applications for this TDD segment are not yet clear, but the spectrum could be useful for domestic or private 3rd generation mobile operators.

With all these constraints, only 2×60 MHz is left to provide full-featured 3rd generation mobile systems and services (and not all of this available globally). This is perhaps sufficient for 2 operators in any one location, each with 2×30 MHz or 3 operators with 2×20 MHz. These figures are not really compatible with the concepts of wideband services and rapid growth being discussed in the context of 3rd generation systems, nor with the current vision of a highly competitive and liberalised network, which would require considerably more than 2 operators to allow real supplier differentiation and consumer choice. Numerous small allocations for operators could also lead to poorer spectrum efficiency. However, until the specific spectrum needs for practical mobile multimedia services are determined the actual number of operators that could co-exist cannot realistically be determined.

5.3 New influences on 3rd generation spectrum

In the previous sections it was shown that there are a large number of fora involved in spectrum management. In the context of European 3rd generation mobile additional actors have come on stage to add further contributions to the discussion.

Among these new players are:

- the Universal Mobile Telecommunications System, (UMTS) task force, and its successor the UMTS forum, a major European grouping (promoted by the EC) considering all aspects of 3rd generation mobile, including (as its highest priority) the spectrum expansion and licensing issues,
- the MoU PMT (Memorandum of Understanding, Personal Mobile Telecommunications), a group of German origin, now 'European', which is looking at an interim generation of mobile networks providing essentially 3rd generation services over 2nd generation mobile networks (i.e. GSM),

- 3GIG (3rd Generation Interest Group), a spin-off from the highly successful GSM MoU group that now has a world-wide membership of 140 operators,
- 3GMG (3rd Generation Mobile (Group)), a UK national advisory group on 3rd generation mobile,
- FAMOUS (future advanced mobile universal systems), a trilateral discussion forum of the EC, Japan and the USA looking at the prospects for global harmonisation of 3rd generation spectrum and standards.

Each of these groups will have opinions on 3rd generation spectrum requirements, both in terms of the total requirement and on how the spectrum should be sub-divided in order to deliver the eventual 3rd generation services.

As a guide to what may be required, the UMTS Task Force recently published its final report [9]. The Task Force views 3rd generation systems as providing high-quality wide bandwidth mobile services, and within its pages was a conclusion that some 2×180 MHz of additional spectrum below 3 GHz would be needed before the year 2005 in order to fulfil the UMTS vision of a true 3rd generation mobile environment.

How a challenging requirement of this magnitude might be met is difficult to see at present. Furthermore, this situation must be viewed against the backdrop of the recent CEPT studies of 29.7 MHz to 960 MHz (i.e. DSI phase 2) and 960 MHz to 3.4 GHz (within CEPT-FM) which found harmonised usage for all the spectrum below 3 GHz without allowances for additional UMTS spectrum requirements. This emphasises the great difficulties of trying to build a long-term harmonised frequency table for Europe in such a dynamic development environment. It is possible that, to achieve the 3rd generation vision, the European frequency table will have to be looked at once more.

5.4 UK 2 GHz spectrum developments

Of late the UK 2 GHz spectrum situation has been fairly dormant. With the co-operation of BT, the UK radiocommunications agency managed to clear both the DECT and the DCS1800 bands (see Fig 3) (1710—1784.7 MHz paired with 1815.5—1879.9 MHz) of fixed microwave links at an early stage. As a consequence the personal communications networks (PCN) were able to get up-and-running in the mid-1990s with well-publicised success.

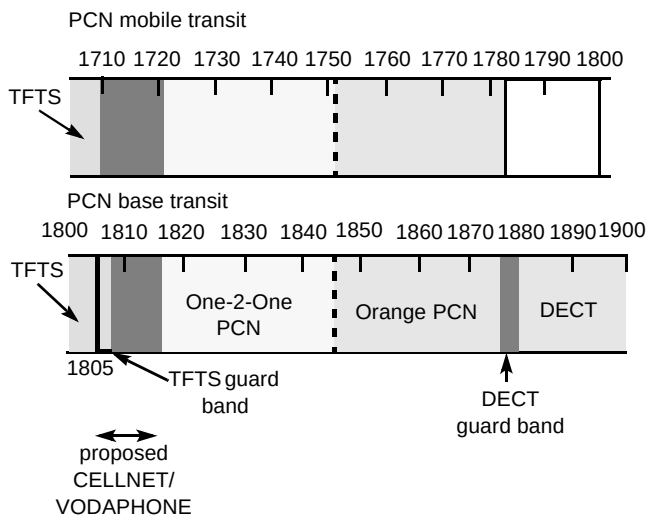


Fig 3 Recent DTI proposals for the DCS1800 band.

Despite three PCN operators being originally envisaged by the UK government only two networks finally emerged. These were each initially allocated 2×15 MHz with a reserve of an additional 2×10 MHz each for possible expansion being held by the UK administration. This left almost 25 MHz unused.

As a welcome step towards providing the four UK mobile operators with more equitable total spectrum allocations, the UK Department of Trade and Industry (DTI) issued a consultation document [10] in February 1996 proposing further developments in the PCN bands. The document proposed that the two existing PCN networks, Mercury One-2-One and Orange, be given extended spectrum allocation of up to 2×30 MHz each (subject to adequate need being demonstrated). Furthermore, it was proposed that up to 2×11.5 MHz could be shared between Cellnet and Vodafone to allow them to develop their respective digital networks. The 11.5 MHz would need to embrace suitable guard band arrangements to protect the Terrestrial Flight Telecommunications System (TFTS) in the adjacent bands. Allowing for a necessary guard-band between the PCN base-station transmitters and DECT, this new package now accounts for all the UK PCN spectrum.

The results of the DTI consultation process on these interesting new proposals are not known at the present time. If these proposed changes are eventually confirmed, then the UK operators will have a clear and stable spectrum environment at 1800 MHz as well as 900 MHz for the foreseeable future. This will greatly help future investment decisions.

6. Conclusions

As this paper has shown, in the three principal areas of interest in mobile spectrum there is considerable ongoing activity.

- At 900 MHz the CEPT activities of the 2nd phase DSI and the PT-25 project team should result in a rationalised and simplified 900 MHz frequency table. This would be a positive step forward as it would provide the operators with a simpler and more stable frequency environment within which to develop their networks.
- At around 2 GHz there is intense activity in a number of fora which hopefully will lead to a utilisation plan and practical transitional arrangements for the 3rd generation mobile network spectrum. However, the EC-supported UMTS task force report has called for an additional 2×180 MHz below 3 GHz before the year 2005 in order to realise UMTS, the European vision of 3rd generation mobile networks and services. Whether this 2×180 MHz is the right figure or not, it does appear that some additional 3rd generation mobile spectrum will need to be found.
- In the UK, interesting proposals have been made by Government relating to further development of the DCS-1800 frequency bands used by the PCN network operators.

These are just the changes affecting the public access mobile networks. Given that there are similar initiatives and issues impacting on PMR, broadcasting and the terrestrial and satellite fixed services (e.g. the issues surrounding the proposed TELEDESIC low Earth orbit satellite network), it is clearly an interesting and challenging time for those involved in spectrum management.

Appendix

The ITU Frequency Table

(Article 8 of the ITU Radio Regulations)

The principal tool employed for managing the frequency spectrum is the Frequency Table. The table is divided vertically (where appropriate) to allow the allocations for each of the three ITU regions (see Fig A.1) to be clearly visible, and it comprises a set of frequency bands, each covering a small fraction of the overall radio spectrum, against which are allocated one or more radio services that can be operated within that band. Figure A.2 shows an example of one page of the definitive global frequency table provided within Article 8 of the ITU radio regulations (in this case as updated in the final acts of WRC-95), this particular example covering some of the important mobile bands at around 900 MHz. The example in Fig A.2 is useful as it shows most of the important features of the ITU frequency table.

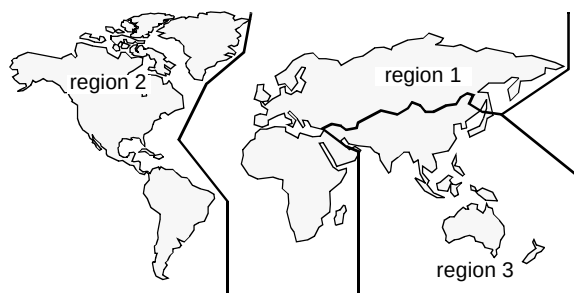


Fig A.1 The three ITU regions.

The frequency range covered by each band is shown, and the bands can be different within each region. Within the bands, the particular service(s) allocated to that band are indicated. Again, these can differ from region to region, as in this example. Services are listed in alphabetical order (according to the French language) and the order does not imply relative priority. The allocations in the Article 8 Table can be of two types.

- Primary allocations, indicated by the use of all capital letters — to ensure an acceptable interference environment, frequencies for new stations of primary services

Allocation to Services		
Region 1	Region 2	Region 3
890 - 942 FIXED MOBILE except aeronautical mobile BROADCASTING S5.322 Radiolocation S5.323	890 - 902 FIXED MOBILE except aeronautical mobile BROADCASTING S5.322 Radiolocation S5.318 S5.325 902 - 928 FIXED Amateur MOBILE except aeronautical mobile Radiolocation S5.150 S5.325 S5.327 928 - 942 FIXED MOBILE except aeronautical mobile Radiolocation S5.325	890 - 942 FIXED MOBILE BROADCASTING Radiolocation S5.326
942 - 960 FIXED MOBILE except aeronautical mobile BROADCASTING S5.322 S5.323	928 - 942 FIXED MOBILE	942 - 960 FIXED MOBILE BROADCASTING S5.320
960 - 1215	AERONAUTICAL RADIONAVIGATION S5.328	
1215 - 1240	RADIOLOCATION RADIONAVIGATION SATELLITE (space-to-Earth) S5.329 S5.330 S5.331 S5.333	

Example footnote:-

S5.322 In region 1, in the band 862 - 960MHz, stations of the Broadcasting Service shall be operated only in the African Broadcasting Area (see Nos. **S5.10** to **S5.13**) excluding Algeria, Egypt, Spain, Libya and Morocco, subject to agreement obtained under No. S9.21.

Fig A.2 Typical page from Article 8 of ITU Radio Regulations showing the principal features.

are generally assigned after being verified against those of existing primary service installations via the process of frequency co-ordination. For a major earth station, for instance, the area of search for potential interference problems (the co-ordination area) can extend out over several hundred kilometres, and can thus cross national boundaries. Frequency co-ordination is thus an international as well as national problem, hence the need for effective co-ordination procedures and the associated criteria and system data to be formally set down in the radio regulations 'treaty'.

Where two or more primary allocations exist in a frequency band (as in the example), then they are deemed to have co-primary status and have equal protection rights on a first-come first-served basis.

- Secondary allocations are indicated by the use of normal characters. Stations of secondary services must not cause harmful interference to existing or future stations of primary or permitted services. Furthermore, stations of a secondary service cannot claim protection from interference from stations of existing or future primary services. They can, however, claim protection from other stations of secondary services which are assigned frequencies at a later date.

Because of the inability to control the interference environment (and hence channel quality), and the possibility of closure if there are problems with protected stations, secondary allocations are of strictly limited value to the telecommunications business.

It should be noted that the allocations in the table can be conditioned in terms of exceptions to 'generic' allocations and, particularly in the case of satellite services, the direction of transmissions.

The frequency table would be simple to implement and manage if it were to comprise only the frequency bands and allocated services for the three regions. However, almost every frequency band, and indeed many individual allocations, are subject to conditional variations contained in numbered 'Footnotes'.

Footnotes are a political/regulatory device which allows the frequency table allocations to be modified where the general case does not meet the requirements of, for example, particular administrations. A Footnote entry in the Article 8 Table provides the relevant administration(s) with a mandate to implement departures from the provisions of the main frequency table within their own borders. Footnotes may be used by administrations to secure:

- additional allocations, i.e. extra services that may be operated in the frequency band in the country or countries mentioned,

- alternative allocations, which replace one or more of the allocations in the main table,
- conditional allocations, which can be brought into force provided that harmful interference will not be caused, e.g. to a neighbouring country or region,
- timetable constraints, whereby an allocation can be caused to enter into force, or perhaps be ceased, on a specified date in certain countries or regions.

The use of footnotes has grown substantially over the years and there are now in excess of 1000 such notes attached to the Article 8 Frequency Table. Individual footnotes can be invoked within several different frequency bands where they are of common relevance. Many of them are sophisticated in their intent and those who use the table must become familiar with their meanings and purposes. An example footnote (No S5.322) is shown in Fig A.2. This particular example recognises the difficulty (indeed practical impossibility) of sharing frequencies between broadcasting and mobile services within the same area, and seeks to constrain the use of the broadcasting service allocation to areas where sharing can be demonstrated not to be a problem. Note the cross reference to other footnotes (Nos. S5.10 to S5.13) which legally define the boundaries of the 'African Broadcasting Area', and to article S9.21, which sets out the frequency co-ordination procedure to be followed to ensure compatibility between systems deployed under the provisions of the footnote allocations and those deployed in neighbouring countries within the mandate of the main table.

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